

Tutorial - Week 11

TG 11 - TG 12

1. Find a formula for the general term a_n of the sequence, assuming that the pattern of the first few terms continues.

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| (a) $\{\frac{1}{3}, \frac{1}{6}, \frac{1}{9}, \frac{1}{12}, \dots\}$. | (d) $\{1, 3, 7, 15, 31, \dots\}$. |
| (b) $\{1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \dots\}$. | (e) $\{1, 2, 6, 24, 120, \dots\}$. |
| (c) $\{5, 8, 11, 14, 17, \dots\}$. | (f) $\{2, 2, \frac{4}{3}, \frac{2}{3}, \frac{8}{15}, \frac{16}{45}, \dots\}$. |

2. Give an example of sequences $\{a_n\}$ and $\{b_n\}$ so that

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n = 0$$

and

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| (a) $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$. | (c) $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = -\infty$. |
| (b) $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \infty$. | (d) $\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$ does not exist. |

3. Determine whether the sequence converges or diverges. If it converges, find the limit.

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| (a) $\{\frac{3 \cdot 2^n}{4n}\}_{n \in \mathbb{N}}$. | (g) $\{\frac{2n-8}{n^2}\}_{n \in \mathbb{N}}$. |
| (b) $\{\frac{1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-1)}{n!}\}_{n \in \mathbb{N}}$. | (h) $\{\cos(\pi n)\}_{n \in \mathbb{N}}$. |
| (c) $\left\{\left(\frac{2n^2}{3n+1}\right)^{\frac{3n^2+2}{5n-3}}\right\}_{n \in \mathbb{N}}$. | (i) $\{\cos(\frac{\pi}{n})\}_{n \in \mathbb{N}}$. |
| (d) $\left\{\left(\frac{2n^2}{3n+1}\right)^{-\frac{3n^2+2}{5n-3}}\right\}_{n \in \mathbb{N}}$. | (j) $\{\cos(\pi n) \cos(\frac{\pi}{n})\}_{n \in \mathbb{N}}$. |
| (e) $\{(\frac{3n-2}{3n})^n\}_{n \in \mathbb{N}}$. | (k) $\{\frac{a^n+b^n}{a^n-b^n}\}_{n \in \mathbb{N}}$ with $0 < a < b$. |
| (f) $\left\{\left(\frac{2n+1}{2n+4}\right)^{\frac{n^2}{n+1}}\right\}_{n \in \mathbb{N}}$. | (l) $\{\frac{na^n+b^n}{a^n-b^n}\}_{n \in \mathbb{N}}$ with $0 < a < b$. |
| | (m) $\{n^k r^n\}_{n \in \mathbb{N}}$ with $k \in \mathbb{N}$ and $ r < 1$. |

4. Study monotonicity and convergence of the sequences given by the following recurrence relations.

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| (a) $\begin{cases} a_{n+1} = \sqrt{2 + a_n}, \\ a_1 = 1. \end{cases}$ | (b) $\begin{cases} a_{n+1} = \sqrt{2 + 2a_n^2}, \\ a_1 = 1. \end{cases}$ |
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5. (a) Let $\{a_n\}_{n \in \mathbb{N}}$ be a decreasing sequence of positive numbers. Then $\sum_{n=0}^{\infty} 2^n a_{2^n}$ is convergent if and only if $\sum_{n=1}^{\infty} a_n$ is convergent.

- (b) Prove that $\sum_{i=1}^{\infty} \frac{1}{n} = \infty$.

6. If a is non-negative, show that $\sqrt{a \sqrt{a \sqrt{a \sqrt{a \dots}}}} = a$.

7. Analyse the convergence of the following series:

(a) $\sum_{n=1}^{\infty} \frac{3+2n}{2-3n}.$

(b) $\sum_{n=1}^{\infty} \frac{n^2}{n^2-2n+5}.$

(c) $\sum_{n=1}^{\infty} \frac{2n-8}{n^2}.$

(d) $\sum_{n=1}^{\infty} \frac{n}{(n+1)(n+2)(n+3)}.$

(e) $\sum_{n=1}^{\infty} 2^{n-1}5^{-n}.$

(f) $\sum_{n=0}^{\infty} [(-0.3)^n - (0.7)^{n+1}].$

(g) $\sum_{n=1}^{\infty} \frac{1}{2+e^{-n}}.$

(h) $\sum_{n=1}^{\infty} \frac{2^n+4^n}{e^n}.$

(i) $\sum_{n=1}^{\infty} (\sin 100)^n.$

(j) $\sum_{n=1}^{\infty} \frac{1}{1+\left(\frac{2}{3}\right)^n}.$

(k) $\sum_{n=1}^{\infty} \ln \left(\frac{n^2+1}{2n^2+1} \right).$

(l) $\sum_{n=1}^{\infty} \cos \left(\frac{\pi}{n} \right).$

(m) $\sum_{n=1}^{\infty} \frac{\sin^n k}{3^n}$ with $k \in \mathbb{R}.$

(n) $\sum_{n=1}^{\infty} ne^{-n^2}.$

(o) $\sum_{n=1}^{\infty} \frac{1}{\sqrt[5]{n^4}}.$

(p) $\sum_{n=1}^{\infty} \frac{\ln n}{n^2}.$

(q) $\sum_{n=1}^{\infty} \ln \left(n \sin \left(\frac{1}{n} \right) \right).$

(r) $\sum_{n=1}^{\infty} \frac{1}{n^n}.$

(s) $\sum_{n=1}^{\infty} \frac{n!}{n^n}.$

(t) $\sum_{n=1}^{\infty} e^{\frac{1}{\sqrt{n}}}.$

8. (a) Approximate the series $\sum_{n=1}^{\infty} \frac{1}{n^3}$ by using the sum of the first 5 terms. Estimate the error involved in this approximation.

(b) How many terms are required to ensure that the partial sum s_n is exactly within $\frac{10^{-6}}{2}$?